

SILAS KWABLA GAH

Complete List of Publications, Manuscripts, and Scientific Preprints

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This document provides a comprehensive, verified record of scientific publications, conference submissions, and targeted journal manuscripts authored by **Silas Kwabla Gah**, accompanying the application for the **Postdoctoral Fellow – Generative AI Researcher – 3D Defect Synthesis** position within the Physical Science and Engineering (PSE) Division at KAUST.

1. Flagship Conference Publications and Active Submissions

Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation

Authors: Silas Kwabla Gah and Ebenezer Owusu [First Author]

Venue / Status: Intl. Conference on Learning Representations (ICLR 2027 Submission; Preprint Available)

Significance / Ranking: CORE A* Flagship Conference (Google Scholar h5-index 300+, Scopus Top 1% globally in AI/ML).

Conducts matched causal interventions across 312 3D indoor scenes (~50M points on ScanNet200 and ScanNet++) to uncover premature decision collapse under greedy hard argmax selection. Proves that retaining compact top-k candidate distributions allows downstream geometric consensus to recover discarded novel categories, boosting 3D segmentation harmonic mean by +6.40 points without parameter fine-tuning.

Training-Free Open-Vocabulary 3D Point-Cloud Segmentation on the Generalized Few-Shot Benchmark

Authors: Silas Kwabla Gah and Ebenezer Owusu [First Author]

Venue / Status: Asian Conference on Computer Vision (ACCV 2026, Paper #482; Accepted / Preprint Available)

Significance / Ranking: Tier-1 Biennial CV Conference (Springer Lecture Notes in Computer Science - LNCS; Scopus Top 10%).

Reconciles dense 3D vision-language representations (RegionPLC) with 2D promptable segmenters (SAM 2.1) using projective camera geometry (K, T), ray casting, and non-parametric multi-view spatial consensus. Establishes state-of-the-art harmonic mean across ScanNet200 and ScanNet++ benchmarks.

Training-Free Generalised Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration

Authors: Silas Kwabla Gah, Ebenezer Owusu, et al. [First Author]

Venue / Status: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027, Paper #167; Under Review)

Significance / Ranking: CORE A Flagship CV Conference (IEEE Xplore / Computer Vision Foundation; Scopus Top 10%, h5-index 104).

Formulates inference-time open-vocabulary composition of promptable segmentation proposals with vision-language embeddings and support-set centroids. Conducts systematic ablation of spatial coordinate transformations, establishing aspect-ratio distortion as an underexamined benchmark confound.

Cross-View Agreement as a Label-Free Process Reward for Multimodal Reasoning

Authors: Silas Kwabla Gah and Ebenezer Owusu [First Author]

Venue / Status: Southern African Conference for AI Research (SACAIR 2026 Submission; Revised)

Significance / Ranking: Springer Communications in Computer and Information Science (CCIS Series; Scopus Indexed).

Formulates physical consistency across independent 2D and 3D viewpoints as an intrinsic, label-free process reward for multimodal reasoning models (Qwen2.5-VL-7B). Built an online GRPO/VERL RL training pipeline on vLLM; conducted counterfactual stress-testing to eliminate spatial area-bias specification gaming ($\rho > 0.72 \rightarrow 0.334$) without human reward annotations.

2. Journal Manuscripts and Published Articles

Reliability-Calibrated Adaptive Semantic Arbitration of Frozen Foundation Model Representations

Authors: Silas Kwabla Gah and Ebenezer Owusu

[First Author]

Venue / Status: Target: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) — In Preparation

Significance / Ranking: Scopus 99th Percentile / Premier Global Journal in AI & Computer Vision (Impact Factor 20.8, CiteScore 37.8).

Comprehensive theoretical foundation for non-parametric inference-time model coordination ($\Delta\theta = 0$). Unites Platt confidence calibration with Bernoulli uncertainty estimation to decouple source confidence from contextual reliability. Formally proves class-wise bounded updates and probability simplex preservation, establishing rigorous safety and calibration guarantees for foundation model composition.

Transfer Learning-Based Encoder–Decoder Model for Skin Lesion Segmentation

Authors: J. K. Appati, L. M. Boante, E. Owusu, and Silas Kwabla Gah

[Co-Author]

Venue / Status: Communications in Computer and Information Science (CCIS), Vol. 1841, pp. 117–128, Springer, 2023

Significance / Ranking: Scopus-Indexed Conference Proceedings / Book Series (Springer Nature; DOI: 10.1007/978-3-031-48879-5_10).

Developed a transfer-learning convolutional encoder-decoder framework for automated boundary extraction in dermatological imaging, providing foundational experience in morphological anomaly segmentation and medical boundary delineation.

3. Doctoral Dissertation

Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations

Authors: Silas Kwabla Gah (Supervised by Prof. Ebenezer Owusu)

[Doctoral Thesis]

Venue / Status: Department of Computer Science, University of Ghana, Legon — Defence Scheduled for October 2026

Significance / Ranking: PhD Dissertation in Computer Vision and Machine Learning.

Investigates non-parametric inference-time coordination of heterogeneous frozen visual foundation models, 3D point-cloud and volumetric segmentation under extreme supervision scarcity, mitigation of premature semantic collapse, and reliability-calibrated evidence arbitration.

4. Codebase Availability & Double-Blind Review Protocol

In strict compliance with international conference **double-blind peer-review policies** (for active submissions to ICLR 2027 and WACV 2027), primary research codebases and full training checkpoints are maintained in private repositories during active evaluation.

- **Technical Documentation & Results:** Complete architectural schematics, experimental protocols, and ablation benchmarks are published on my academic portfolio at: silsgah.github.io/#projects.
- **Committee Code Access:** Full repository access, reproduction environments, or an anonymized review branch can be provided immediately to members of the KAUST search committee upon request via gahsilas@gmail.com.